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Unsustainable prosperity? Decoupling wellbeing, economic growth, and greenhouse gas emissions over the past 150 years

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ABSTRACT

Since the Industrial Revolution, modern economic growth has made the world increasingly (if unevenly) rich. This trajectory led to unprecedented improvements in human wellbeing but, at the same time, produced environmental impacts which threaten material prosperity itself. Against this background, the great challenge of the 21st century is to continue to improve global wellbeing while mitigating environmental impacts. For most international organizations and scholars this can be achieved by decoupling economic growth from its environmental costs (as in green growth proposals). However, other scholars question the feasibility of sustaining growth with a sufficiently large fall in its ecological footprint, arguing instead for decoupling wellbeing from economic growth (as in post-growth strategies). How common were these two forms of decoupling in the past? When and where did societies manage to separate wellbeing improvements from environmental impacts? The answers to these questions can allow us to trace the long-run direction of travel of the global economy and to rethink the historical narrative of the emergence and consolidation of modern economic growth by considering its impacts on human wellbeing and environmental change at the same time. We offer the first long-term analysis of decoupling patterns between an augmented human development index (AHDI) and greenhouse gas emissions (GHGe). Moreover, we identify when these patterns were explained by 'growth decoupling' (GDP grows faster than GHGe), by 'wellbeing decoupling' (AHDI grows faster than GDP per capita), by both, or by neither. Our results show that at low income levels all world regions experienced episodes of wellbeing decoupling; as they became richer, growth decoupling became more common. Nevertheless, we find that such decoupling episodes have proved reversible and that no country in the world has yet managed to achieve very high levels of human wellbeing within planetary boundaries.

> *(T)he industrial revolution may come to be regarded not as a beneficial event which liberated mankind from the shackles which limited growth possibilities in all organic economies but as the precursor of an overwhelming tragedy – assuming that there are still survivors to tell the tale.* (Wrigley, 2016, p. 205)

1. Introduction

After centuries of Malthusian dynamics during which economic growth was unable to provide sustained income increases for rising populations, the Industrial Revolution unleashed an era of unprecedented prosperity which eventually reached all world regions (Allen, 2011; Koyama & Rubin, 2022). The persistent growth of output per person (measured as GDP per capita) allowed for worldwide —albeit unequal— improvements in wellbeing (Deaton, 2013), but required ever more natural resources and produced increasingly large environmental impacts (Krausmann et al., 2009; Rockström et al., 2023; Steffen et al., 2015). The acceleration of the environmental impacts of economic activity threatens human wellbeing and planetary health. Six of the nine so-called 'planetary boundaries' to maintain the stability of the Earth system have been surpassed (Rockström et al., 2023). Modern economic growth seems, in consequence, trapped in a paradox: it improves and at the same time threatens the human condition. The world economy has relied for over two centuries on dominating and reshaping nature, while the natural environment increasingly leaves its mark on our

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economies—like sandalwood perfuming the axe that lays it low, to borrow Rabindranath Tagore's verse.[1]

Against this background, decoupling economic growth from its environmental impacts ('growth − impact decoupling') has become a key concept in the academic debate (Haberl et al., 2020; Jackson & Victor, 2019) as well as in the global policy arena (OECD, 2011; UNEP, 2011a; World Bank, 2011) as one of the Sustainable Development Goals. By comparison, much less attention has been paid to decoupling human wellbeing from environmental impacts (exceptions are Akizu-Gardoki et al., 2018; Hossain & Chen, 2021; see: Li et al., 2022). This is a critical omission because increasing output per capita is ultimately a means to an end —improving wellbeing— and, as the 'beyond GDP' literature continues to show, GDP per capita is a very limited proxy for wellbeing (Costanza et al., 2014; Hoekstra, 2019; Stiglitz et al., 2009). In fact, an increasing number of scholars argue for the possibility of improving human wellbeing while limiting economic growth, and thus, its environmental impacts (Jackson, 2021; Raworth, 2017; Roberts et al., 2020). Inspired by the 'Oxfam doughnut' (Raworth, 2017), recent studies analyse whether countries manage to achieve sufficiently high levels of wellbeing (measured by a range of subjective indicators and need satisfiers) within planetary boundaries, or, in other words, whether countries can find a place within a 'safe and just operating space' (Dearing et al., 2014; Fanning et al., 2022; O'Neill et al., 2018; Rockström et al., 2023). None of these studies, however, has studied the *long-term* relationships between wellbeing, economic growth, and environmental impacts as we aim to do in this paper.

These relationships have long been overlooked by historical scholarship too. Economic historians have mostly focused on the evolution of incomes in the long-run, studying the trends and discontinuities of modern economic growth and showing both the overall improvements in living standards as well as the rise of income inequalities (Allen, 2001; Chancel & Piketty, 2021; Deaton, 2013; Pomeranz, 2000). In the last 200 years, global inequality in living standards has been driven primarily by location rather than by class (Milanovic, 2013). This explosion of between-country inequality is read by some economic anthropologists and sociologists as evidence of persistent (or re-emerging) divides driven by unequal ecological exchange in the context of Western colonialism and neo-colonialism (Hickel, 2017; Savage, 2021), whereas most development economists and economic historians emphasise the substantial improvements in living standards across the world since industrialization, also in developing regions and including periods under colonial rule (Gardner & Roy, 2020; Koyama & Rubin, 2022).

Only recently have historians started to systematically include other indicators of material and social wellbeing which have revealed that, while correlated, income and wellbeing do not always have paired trajectories: divergence and convergence across world regions or within them look different when broader human development indicators are considered (Prados de la Escosura, 2022a; van Zanden et al., 2021a). But economic historians have yet to include environmental metrics in these studies. Meanwhile, most environmental histories do not systematically consider the relationship of environmental impacts with social or economic indicators. Only a handful of studies have analysed the long-term relationships between resource use and GDP growth (Gales et al., 2007; Kander et al., 2014), or between carbon dioxide emissions and GDP (Henriques & Borowiecki, 2017; Lindmark, 2004), but without considering human wellbeing as such. Meanwhile, several studies of long-term socio-ecological transitions have examined the relationship between economic growth and natural resource use at the global level (Bithas & Kalimeris, 2018; Haas et al., 2020; Krausmann et al., 2009), as well as for specific countries, such as the United States, (Gierlinger & Krausmann, 2012), the USSR (Krausmann et al., 2016), the United Kingdom (Krausmann et al., 2008), France (Magalhães et al., 2019), or Spain (Infante-Amate et al., 2015). Yet, this strand of literature does not link biophysical metrics with human wellbeing indicators.

In this paper we examine for the first time the decoupling patterns between an augmented human development index (AHDI), GDP, and greenhouse gas emissions (GHGe) since 1870, globally and regionally. In so doing, we aim to (i) examine global and regional trends of decoupling wellbeing from environmental impacts; (ii) identify whether this was due to the decoupling of wellbeing improvements from economic growth, or to the decoupling between economic growth and emissions; (iii) analyse in which historical settings these different decoupling episodes have taken place; (iv) consider whether regional development trajectories have ever reached a 'safe and just operating space'.

After this introduction, section 2 reviews the scholarly debate regarding the viability of decoupling and the specialist literature on decoupling measurement. Section 3 presents our data sources and approach. Section 4 offers the main results on decoupling and divergence patterns in world and regional development since 1870. In section 5 we discuss these results, considering their implications for the current debates surrounding 'green growth' and for the historical interpretation of global economic development. The last section offers our conclusions.

2. Decoupling: Views and concepts

There are broadly speaking two major camps when it comes to decoupling studies: those who view green growth as possible and desirable (Figure 1a) and those who are sceptical about its prospects and consider instead that it is human wellbeing which must be decoupled from output per capita (Figure 1b). Among the first group, we find the international organizations which promote green growth as a core policy goal and thus must by definition consider it possible despite recurrent failures to meet national-level targets (e.g. the European Union with its European Green Deal, the United Nations with its Sustainable Development Goals, and even before the OECD (2011), the World Bank (2011) and other programs such as UNEP (2011b)). The academic backbone of green growth proposals is the hypothesis known as the Environmental Kuznets Curve (EKC, reviewed extensively in Dasgupta et al. (2002), Dinda (2004) and Stern (2004). The EKC –inspired by Kuznets' (1955) speculation about the inverted-U shape relationship between economic growth and inequality in the long-run– posits that environmental impacts tend to increase sharply with industrialization but that after a certain turning point they start to decrease. If this was true, then economic growth would be part of the solution to environmental sustainability challenges, rather than part of the problem. The mechanisms implied are that as economies develop, technological change allows for a reduction in environmental impacts per unit of output, and structural change increasingly shifts output towards less energy-intensive and less polluting industries (Grossman & Krueger, 1991; Panayotou, 1993; Riahi et al., 2017; van Vuuren et al., 2017). The microeconomic foundations for these changes could be found in the income elasticity of quality demand: higher-income individuals will be willing to pay a premium for environmental protection, which will encourage R&D and public policy efforts towards impact mitigation (see the review by Dinda (2004)).

Nevertheless, for many scholars, sustained economic growth could be incompatible with an *absolute* mitigation of environmental impacts, or, even instances of such 'absolute decoupling' were possible, the scale of mitigation would be insufficient to keep the global economy within planetary boundaries (e.g. Hickel & Kallis, 2020; Vogel & Hickel, 2023; Wiedmann et al., 2020). They argue that however impressive technological innovation and structural change have been in recent history, they have proven insufficient to compensate the overall expansion of economic output. While sceptical about green growth, the proponents of academic or political approaches such as degrowth (e.g. Hickel et al., 2022; Kallis et al., 2012), agrowth (Van den Bergh, 2011), post-growth (Jackson, 2021), or doughnut economics (Raworth, 2017) claim that

[1] The Bengali poet is widely credited as the author of the verse "As if to prove that love would conquer hate, the sandalwood perfumes the very axe that lays it low" (see for example Pearlstine, 2022, p.40).

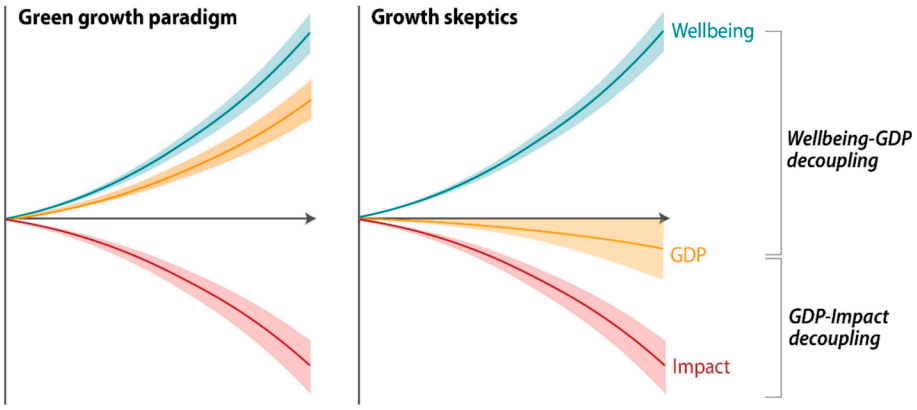


Figure 1. Stylised representation of impact decoupling and wellbeing decoupling in the green growth debate.

limiting economic growth need not lead to a reduction of human wellbeing. On the contrary, they posit that through redistributive policies, targeted growth restrictions in richer countries, or controls on specific sectors of the economy, it would be possible to increase global wellbeing without expanding overall output. In other words, they consider more feasible to improve human wellbeing and avoid global environmental catastrophe by decoupling GDP per capita from wellbeing, rather than GDP per capita from its environmental impacts.

In sum, green growth advocates claim that to avoid the negative impacts associated to global environmental change it is sufficient to keep the economy growing without causing more environmental degradation. This would be achieved through 'growth – impact decoupling': dissociating GDP increases from environmental impact thanks to technological innovation and structural shifts in the economy. Green growth sceptics, on the other hand, consider this kind of decoupling unfeasible or extremely insufficient even under the most optimistic projections, and propose instead policies and actions intended to reduce or stabilize economic output while decoupling it from human wellbeing. Despite this apparently unbridgeable divide, both camps share the ultimate goal of improving global human wellbeing while fundamentally mitigating environmental impacts. They differ in how to get there: the former rely on technological innovation to decouple production from resource extraction and pollution; the latter trust socio-political innovation that would separate improvements to living standards from economic growth. It should be said that some green growth advocates do also consider there is substantial margin for wellbeing decoupling. Joseph Stiglitz, who has recently criticised growth sceptics (Stern & Stiglitz, 2023), led an influential report arguing for broadening the focus of policy-making beyond the sole goal of GDP growth, considering alongside it a range of wellbeing indicators (Stiglitz et al., 2009). Meanwhile, green growth sceptics are by no means opposed to mitigation through technological change, which they consider important but insufficient to avoid further climatic and environmental crises. Growth decoupling and wellbeing decoupling are of course not mutually exclusive, and, as we will show below, both have important historical precedents.

3. Definitions, Data, and methods

3.1. Approaches to decoupling

Decoupling can be studied in different ways depending on the choice of indicators, the approach to examine their relationships, and the scale of the study. Here we briefly consider some of the main approaches in the scholarship before setting out our own in the next section.

Firstly, the decoupling literature distinguishes between absolute decoupling and relative decoupling (Haberl et al., 2020; Parrique et al., 2019). Absolute decoupling takes place when GDP per capita increases, while resource use and environmental impacts fall in absolute terms. Relative decoupling occurs when GDP per capita and resource use and environmental impacts all increase, but GDP per capita does so at a faster rate. In this case, environmental efficiency improves, but overall resource use and impacts still increase.

Secondly, decoupling studies vary depending on the variables considered. The most common choice is to examine decoupling between GDP growth and environmental impacts (known as impact decoupling). Many other studies consider instead decoupling between GDP growth and resource use (known as resource decoupling). Only a few studies have analysed decoupling between wellbeing metrics and environmental impacts (Hossain & Chen, 2021; Li et al., 2022), or wellbeing and resource use (Akizu-Gardoki et al., 2018; Kalimeris et al., 2020).

A third distinction lies in the way these variables are quantified. GDP estimates follow an internationally harmonized system of national accounts. Measuring environmental impacts in the context of decoupling research is more complicated. Environmental impacts are multidimensional, and each dimension comes with its own metrics. The widely used 'planetary boundaries' framework identifies up to 9 kinds of environmental impact to be quantified in different ways (Rockstrom et al., 2009). While most decoupling studies focus only on CO_2 emissions arising from fossil fuel use, there are analyses which consider other greenhouse gas emissions or even other kinds of impacts altogether, such as biodiversity loss and freshwater use (see the review in Parrique et al. (2019)). Likewise, resource use can also be counted in different ways through material flow accounting, energy consumption, and land use, among others. Finally, indicators of human wellbeing are also controversial. Much like environmental impacts, wellbeing is multidimensional and depends on a range of factors both material and immaterial. An important distinction is between the material conditions of wellbeing, i.e. the 'need satisfiers' that allow people to live better lives – such as access to education, healthcare, and the ability to participate equally in politics – and subjective wellbeing, i.e. the individual perception of experienced happiness, usually measured with survey data (Alkire, 2002; Deaton, 2013; Kahneman & Deaton, 2010).

A fourth way to classify studies is to consider their spatio-temporal scope: decoupling analyses cover very different chronological and geographical ranges. While there are so far no long-term studies (>100 years) like our own, several studies suggest the importance of considering relatively long periods, because decoupling episodes can be reversed (Hubacek et al., 2021; Shuai et al., 2019), as the IPCC reports have pointed out (IPCC, 2022). In the same vein, having a broad enough geographical scope is crucial because it has been observed that decoupling is often a highly localised phenomenon rather than a far-reaching process (Parrique et al., 2019).

3.2. Data sources

The most widely used measure of human wellbeing is the Human

Development Index (HDI) published by the United Nations since 1990. HDI is a summary indicator resulting from the geometric mean of normalized indexes of life expectancy, years of schooling, and per capita incomes. In this paper we take the series by Prados de la Escosura (2022a) who offers historical HDI estimates for most world countries since 1870. His estimates involve, however, two important changes with regards to the UN's standard metric in order to make long-term historical changes more visible. Firstly, Prados de la Escosura applies the so-called Kakwani transformation to life expectancy and education data to deal with the fact that these two components of HDI face natural limits: unlike income, their growth potential has a definable upper bound. The Kakwani transformation gives more weight to improvements in education and life expectancy as they come closer to the upper bound, reflecting that it is more difficult to increase mean years of schooling from say 12 to 13 than from 4 to 5. Secondly, following Amartya Sen's capabilities approach to development, Prados de la Escosura includes an indicator of civil and political freedom: the 'liberal democracy index' from the Varieties of Democracy database (Coppedge et al., 2023). Including this fourth indicator, Prados de la Escosura's index is an Augmented HDI (AHDI). Throughout this paper we use his AHDI series excluding the per capita GDP component, because we want to compare human development metrics with income per capita and with environmental impacts. Prados de la Escosura's database offers global, regional, and national AHDI estimates for the following years: 1870, 1880, 1890, 1900, 1913, 1925, 1929, 1933, 1938, 1950, 1955, 1960, 1965, 1970, 1975, 1980, 1985, 1990, 1995, 2000, 2005, 2010, 2015, and 2020.

Thoughout this paper, we use primarily the AHDI series by Prados de la Escosura (2022a) because, among all published historical HDI estimates, it is the one with the largest spatial and chronological coverage, and because it allows us to interact with the literature in development economics and economic history that relies on it. Nevertheless, Prados de la Escosura's two variations relative to the canonical United Nations' HDI definition—the Kakwani transformation of life expectancy and years of schooling, plus the addition of the 'liberal democracy index'—can result in substantial level changes relative to a more conventional HDI estimate. For example, the UN reports an HDI for present-day China of 0.77 (20 % lower than the highest national HDI in the world), whereas the AHDI gives it a value of 0.18 (80 % lower than the highest AHDI value in the world).[2] To address this issue, we have calculated a UN-style HDI series based on the primary data on GDP per capita, years of schooling, and life expectancy reported by Prados de la Escosura. This means we can replicate our results with a historical HDI that matches the conventional UN series, caluclated as 'the geometric mean of the normalized indices for each of the three original dimensions' (United Nations, 2024). For the indicators of education and health we follow the UN normalization thresholds, but for GDP per capita we take 100 international dollars of 1990 as the lower-bound, and 47,000 as the maximum bound, following Prados as these thresholds reflect better the long-term variation in incomes. Although throughout the article we present most results using the AHDI series, we have replicated the key results and analyses using this conventional historical HDI (with or without the income component, as appropriate) to assess whether the trends we describe and the conclusions we arrive at differ substantially. Figure SM1 shows the historical evolution of both the AHDI and the standard HDI series.

GDP per capita data are taken from the Maddison Project Database (MPD) (Bolt and van Zanden, 2020). This database offers long-run GDP and population estimates for almost all world countries in an unbalanced panel dataset, with some series starting before others. At the level of world regions there are no annual estimates, but a balanced panel of benchmark years between 1820 and 2018. In order to have annual series at the regional level, we calculate regional per capita GDP from national-level estimates when the MPD offers GDP and population data for countries representing more than 80 % of the region. When the national estimates available do not meet this threshold, we keep the MPD's original regional series and interpolate linearly between benchmark years. We further extend the per capita GDP data up to 2020 using the national growth rates reported by OECD (2023).

Finally, data on greenhouse gas emissions is constructed from different databases. Fossil-fuel and cement CO_2 emissions are taken from the Global Carbon Project (GCP), which provides estimates for all world countries since 1750 (Andrew & Peters, 2021). Land-use, land-use change, and forestry (LULUCF) CO_2 emissions are taken from the recent estimate by Jones et al. (2023). Their study offers annual series based on the average of three bookkeeping estimates also used by the GCP (Gasser et al., 2020; Hansis et al., 2015; Houghton & Nassikas, 2017). CH_4, N_2O, and fluorinated gases emissions are taken from the PRIMAP-Hist database (Gütschow et al., 2016; Gütschow & Pflüger, 2023). These include emissions released by the energy sector (fugitive emissions), as well as those stemming from agriculture, forestry, and other land uses (AFOLU) and from other sectors. We rely entirely on production-based estimates, that is, we do not consider the emissions embodied in international trade and, consequently, our analysis does not offer a consumer perspective simply because we currently lack appropriate estimates. Information on greenhouse gas emissions embodied in international trade is only available since 1990 for fossil-fuel CO_2 (Brolin & Kander, 2022; Friedlingstein et al., 2022). Working exclusively with production-based accounting underestimates emissions caused by wealthier countries (and overestimates those by poorer countries), as existing evidence suggests that high-income economies externalise a share of their emissions to other countries, and thus their carbon footprint is larger than their territorial emissions would suggest (Friedlingstein et al., 2022; Peters & Hertwich, 2008; Wiedmann, 2016).

3.3. Indicators

We examine first the evidence of decoupling between income, emissions, and wellbeing indicators. To do so, we compare the trends for each variable, as well as the relationship between all pairs of variables using standard intensity indicators. We consider (i) the carbon intensity of GDP, measured as the GHGe by unit of economic output value; (ii) the wellbeing intensity of economic growth, measured as the improvement in AHDI per unit of GDP per capita; and finally (iii) the carbon intensity of the AHDI, that is, the per capita emissions released per unit of improvement in the wellbeing indicator. We focus on this last relationship and, inspired by the Tapio (2005) model, we identify five patterns of coupling and decoupling (Figure 2a): (i) dirty development, when both AHDI and GHGe grow, but GHGe grow faster; (ii) relative decoupling, when both AHDI and GHGe grow, but AHDI grows faster; (iii) absolute decoupling, when AHDI grows and GHGe fall; (iv) underdevelopment with emission reduction, when both GHGe and AHDI fall; and (v) dirty underdevelopment, when AHDI falls and GHGe grow.

We then turn to the components of wellbeing-impact decoupling in order to address current development debates (see Figure 1). We decompose wellbeing-impact decoupling in wellbeing-income decoupling (comparing AHDI and GDPpc) and growth-impact decoupling (comparing GDP with GHGe).[3] The existing methods for decoupling analysis are designed to examine the relationships between two

[2] For more arguments against and in favour of this methodological decisions by Prados, see, respectively, van Zanden (2022) and Prados de la Escosura (2022).

[3] Note that we compare AHDI with GDP per capita: despite the former not being adjusted by population size, these are both indicators of average individual-level standards of living. In the case of growth-impact decoupling we compare total GDP with total GHGe (without normalising by population size) because in terms of environmental impacts what matters are the absolute values. The atmosphere does not care about per capita emissions, but about total emissions.

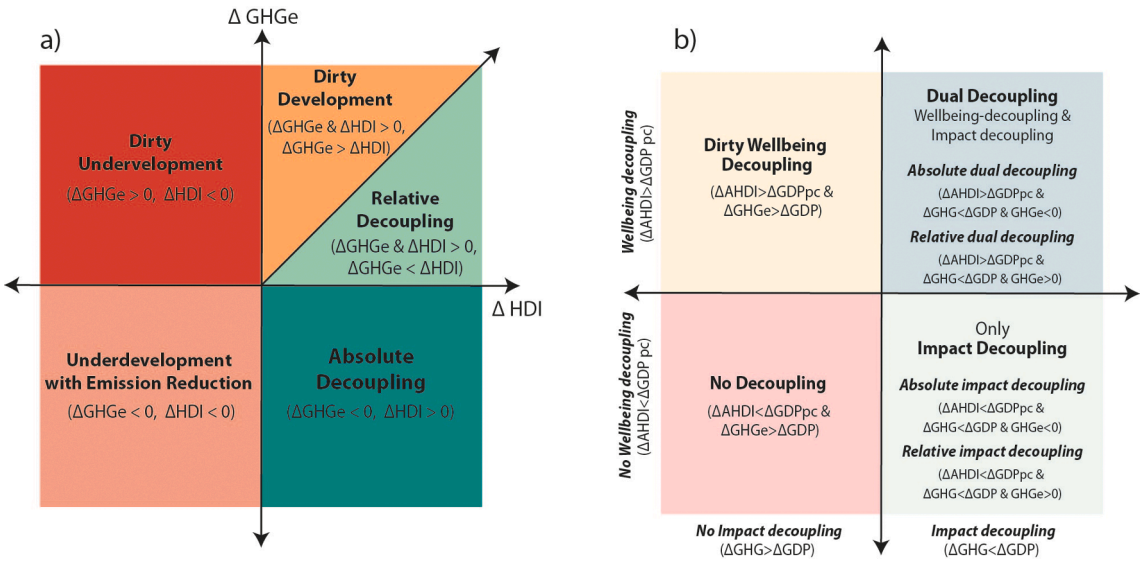


Figure 2. (a) Decoupling patterns between wellbeing and greenhouse gas emissions. (b) Framework with intersections between wellbeing-income decoupling and growth-impact decoupling.

variables (see a review in Wu et al. (2018)). In this case, however, we want to consider two relationships implying three variables: AHDI, GDP, and GHGe. To do so, we propose a simple framework to examine, on the one hand, the comparative evolution of AHDI and GDPpc, and, on the other hand, the comparative evolution of GDP and GHGe. The intersection of these two dimensions yields four combinations, which, when linked with the potential scenarios of 'wellbeing-impact decoupling,' result in twenty possible scenarios. Empirically, however, only 6 combinations account for effectively all historical regional scenarios. Therefore, we now consider the prevalence of these 6 patterns at a global, regional, and national level across different periods (Figure 2b and Table 1). Firstly, dirty wellbeing decoupling, when we find wellbeing decoupling without growth-impact decoupling (top left quadrant). Secondly, growth-impact decoupling without wellbeing-income decoupling (bottom right quadrant), which can take place under increasing emissions in absolute terms (relative impact decoupling) or with falls in emission levels (absolute impact decoupling). Thirdly, both wellbeing-income and growth-impact decoupling occurring simultaneously (top right quadrant). Similarly, this scenario can occur under absolute increases (dual relative decoupling) or absolute falls (dual absolute decoupling) in GHGe. Finally, we consider a scenario where there is no decoupling at all (bottom left quadrant).

Table 1
Wellbeing-impact decoupling patterns according to their components: AHDI-GDPpc decoupling and GDP-GHGe decoupling.

Pattern	*Description*	*Implication*
Absolute dual decoupling	GHGe fall in absolute terms and AHDI increases; AHDI grows faster than GDPpc, and GDP grows faster than GHGe.	The ideal scenario: both 'green growth' and 'post-growth' decoupling standards are met with emission mitigation.
Relative dual decoupling	GHGe grow and AHDI improves; AHDI grows faster than GDPpc, and GDP grows faster than GHGe.	Economic growth and wellbeing improvements but no emission mitigation.
Absolute impact decoupling	GHGe fall in absolute terms and GDP grows; GDPpc grows faster than AHDI.	Economic growth with emission mitigation but slower wellbeing gains.
Relative impact decoupling	GHGe increase but GDP grows faster than GHGe; GDPpc grows faster than AHDI.	Economic growth without emission mitigation and with slower wellbeing gains.
Dirty wellbeing decoupling	GHGe increase but GDP grows faster than GHGe; AHDI grows faster than GDPpc.	Economic growth and faster wellbeing improvements, but environmental impacts increase.
No decoupling	GHGe increase faster than GDP; AHDI grows less than GDPpc.	The worst scenario: economic growth is environmentally inefficient and socially underwhelming.

In the final section of our Results, we examine the levels of per capita emissions and of AHDI at the regional and national levels. We contrast their trajectories against a historical 'safe and just operating space', whose boundaries we define as follows. In terms of justice, we consider that all countries should be able to converge to the average wellbeing standards enjoyed by Western countries since 2000 (AHDI>0.69), although we also compare each observation with the historical global average for AHDI (0.21). In terms of safety for planetary health, we allocate to each country the per capita emissions that they would be entitled to between 1850 and 2050 to maintain global mean temperatures below 1.5 °C above pre-industrial levels (a similar approach to Vogel & Hickel (2023)). To perform this calculation we simply divide the carbon budget available to fulfill this climate agreement among the global population between 1850 and 2050. We take historical populations from the sources cited above and we rely on the United Nations (2022) projections for 2050. The carbon budget is estimated by adding to historical emissions (see above) the remaining carbon budget updated by Friedlingstein et al. (2023) compatible with a 50 % probability of staying below a 1.5 °C increase by 2050. The result is that the historical budget was about 3.5 tons of CO_2e hab^{-}1: that is, if all people had emitted that amount between 1850 and 2050, mean temperatures would remain below 1.5 °C by 2050. Following the same procedure, the safe operating space remaining from our own days until 2050 would be of 0.4 tons of CO_2e hab^{-1}.

4. Results

4.1. Global trends

Wellbeing, income, and greenhouse gas emissions have all increased substantially over the last 150 years. Figure 3 shows the global trajectory of wellbeing (proxied by the AHDI without the income component), GDP, and GHGe since 1870: it holds a historical mirror to Figure 1's theoretical illustration. Until the First World War, all 3 variables were coupled: wellbeing increased at roughly the same pace as per capita GDP, which caused proportional impacts on emission per capita levels

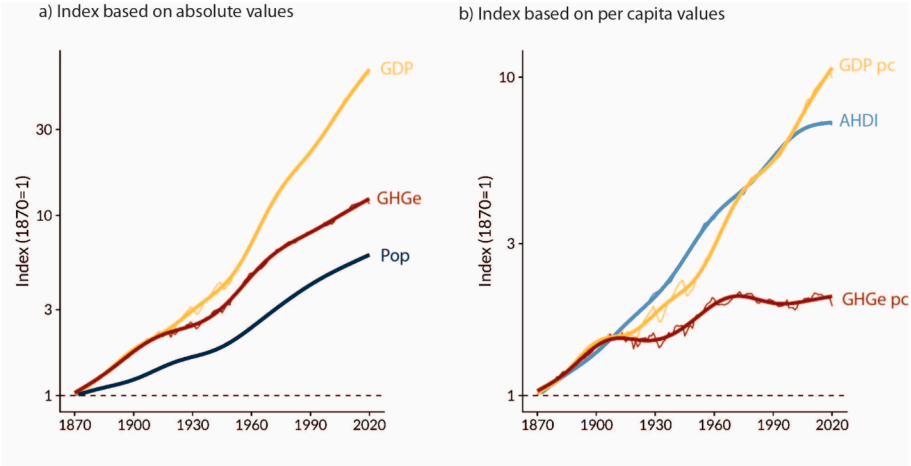


Figure 3. World development since 1870: Population, wellbeing (AHDI, i.e. Augmented Human Development Index without the income component), income (GDP, i.e. Gross Domestic Product), and environmental impact (GHGe, i.e. greenhouse gas emissions) (see Figure SM2 for regional trajectories).

(all growing annually ~ 1.1 %). In absolute terms, similarly, GHGe and GDP also grew in tandem (~1.9 %). We can also examine these trends through the changes in relative intensities. Figure 4 shows that the carbon intensity of GDP (emissions per unit of economic output) remained largely stable between 1870–1913, as well as the carbon intensity of AHDI and the intensity of wellbeing per unit of income.

This pattern changed between 1914 and 1945: wellbeing and economic growth expanded faster than emissions, due primarily to the fall in fossil fuel use in Western countries, especially in war-torn Europe. Considering the world as a whole, this was a period of 'relative dual decoupling' (Table 1 and Table 2). We observe a fall in the carbon intensity of GDP and, at the same time, an increase in the AHDI per unit of GDP per capita. The global economy became overall cleaner in relative terms, while the AHDI grew faster than GDP per capita, even if improvements were much slower than in later periods. In the discussion below we will examine the apparent contradiction of improving wellbeing in a time of formidable social crises, including two world wars and the Great Depression. Given these trends, we observe a sharp fall in emissions per unit of AHDI.

After the Second World War, there was a 'great acceleration' (McNeill & Engelke, 2014; Steffen et al., 2015) in output (annual growth of 3.9 %) alongside its environmental impacts (2.4 %), and the AHDI (1.6 %). All three indicators reached their historical peak growth rates, although the impact of the post-war economic boom was larger on the environment than on wellbeing. During this period, the carbon intensity of GDP kept falling but, for the first time in the series, AHDI grew more slowly than per capita GDP. As a consequence, a process of relative

Table 2
Annual growth rates in selected indicators.

	AHDI	Absolute GDP	GHGe	Per capita GDP pc	GHGe pc
1870 – 1913	1.1	1.9	1.9	1.1	1.1
1913 – 1945	1.6	1.8	0.8	0.9	−0.1
1945 – 1990	1.6	3.9	2.4	2.1	0.6
1990 – 2020	0.8	3.3	1.2	2.0	−0.1
1870 – 2020	1.3	2.8	1.6	1.5	0.4

improvements in environmental efficiency was combined with social inefficiencies, as economic growth did not result in equivalent improvements to wellbeing.

Since 1990, there is substantial evidence of 'relative impact decoupling' (with GDP per capita outpacing emissions) but no evidence of wellbeing decoupling at the global level. This period marked the consolidation of some of the postwar trends: the global growth gap between incomes and wellbeing became even larger, with GDP per capita gains leading to smaller AHDI improvements than before, while GDP per capita has expanded much faster than GHGe. Nevertheless, total emissions have not ceased to increase globally until the present, with the exception of short-livebd historical episodes such as the COVID-19 pandemic.

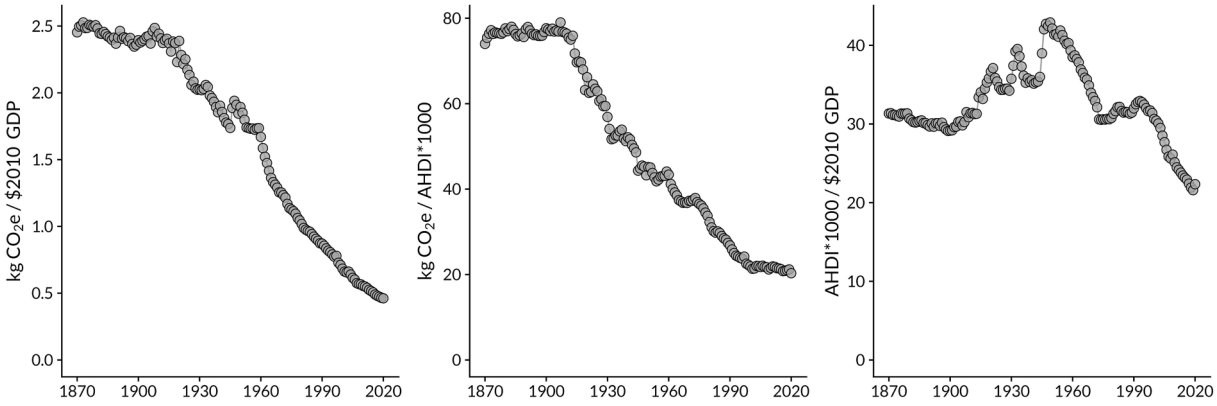


Figure 4. (a) Carbon intensity of GDP, measured as kg of CO2e per $ of GDP. (b) Carbon intensity of AHDI, measured as kg of CO_2e per AHDI multiplied by 1000. (c) Well-being intensity of GDP, measures as AHDI multiplied by 1000 divided by $ of GDP.

4.2. Regional and global decoupling of AHDI and GHGe

Considering first wellbeing-impact decoupling, we find that in the early stages of modern economic growth countries most often went through a process of 'dirty development', with emissions outpacing wellbeing. Episodes of relative decoupling between wellbeing and emissions were exceptional at a regional level, although they have become increasingly recurrent in most world regions since the second half of the 20th century (Figure 5). Nevertheless, this evolution is not linear nor permanent: indeed, it has been reversed in many cases. Between 1913 and 1950 national-level 'relative decoupling' episodes were more common, whereas in the so-called 'golden age of capitalism' following World War II dirty decoupling episodes became widespread, except in sub-Saharan Africa. In the late-20th century relative decoupling became once again generalised, with cases of absolute decoupling in Europe and the United States as well as in Latin America and sub-Saharan Africa (Figure 5).

Let us turn to regional paths (Figure 5b). In sub-Saharan Africa and Latin America, dirty development patterns dominated in the late-19th and early-20th century, as emissions grew faster than wellbeing. Afterwards decoupling (both absolute and relative) has been recurrent. In Latin America wellbeing-impact decoupling was widespread between the 1930s and the 1980s; dirty development has become more common since. Decoupling patterns in Sub-Saharan Africa ran contrary to the global average during most of the 20th century. In the interwar period, when most countries show evidence of relative decoupling (Figure 5a), sub-Saharan Africa experienced dirty development, led by fast-increasing land-based emissions (Figure SM3). Conversely, in the 'golden age of capitalism', African countries experienced both relative and absolute decoupling episodes, while dirty growth dominated the rest of the world. Once again, the relative or absolute fall in GHGe in Africa was related to changes in land-based emissions due to the slowdown of deforestation processes.

In Western countries (Western Europe and Western Offshoots), earlier stages of modern economic growth were also characterised by dirty development (Figure 5). Intermittent episodes of relative and absolute decoupling show up in the early-20th century, even more so in the 'golden age' of post-war economic growth, and in the 21st century. In the so-called 'Western Offshoots' (the United States, Canada, Australia, and New Zealand) we find that since 2010 emission mitigation was accompanied by a fall in AHDI. Even if GDP has grown, the non-income components of wellbeing seem to be decreasing. However, the fall in

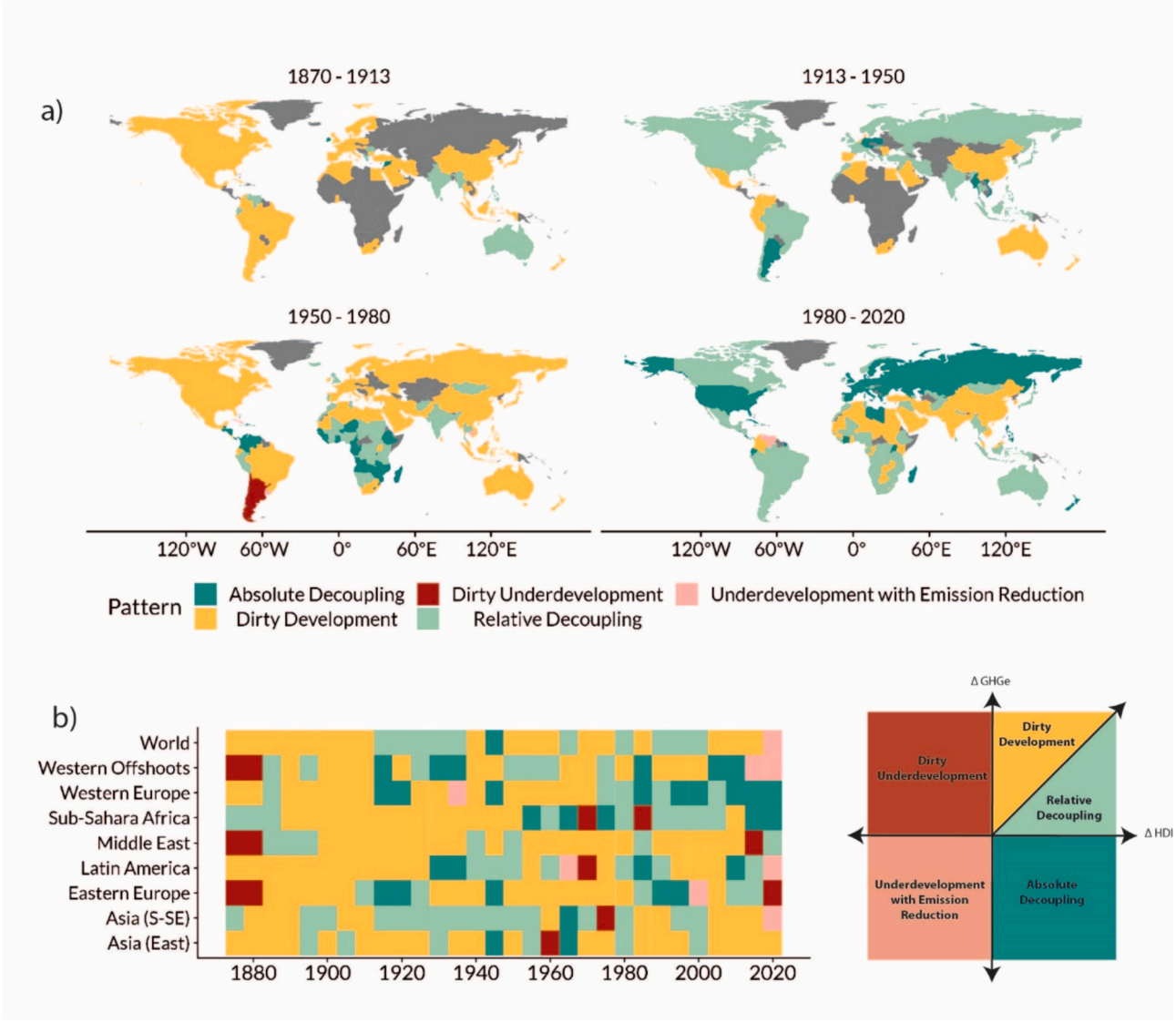


Figure 5. Tapio patterns linking variations in AHDI (Augmented Human Development Index) and greenhouse gas emissions (GHGe) as per Figure 1. Figure SM4 replicates this figure using the conventional HDI (without the income component).

wellbeing is not observed if we consider the conventional HDI (without the income component) instead of the AHDI (see Figure SM4). This phenomenon of 'underdevelopment with emission reduction' had already occurred in interwar Europe and has re-emerged in the West and in other regions between 2015 and 2020 due to the effects of the coronavirus pandemic.

Asian economies, albeit with much variation across them, have been broadly characterised by intermittent switches between dirty development and relative decoupling patterns (Figure 5). During the first decades covered by our study, dirty growth dominated although there were several episodes of relative decoupling. For most of the 20th century relative decoupling prevailed, although dirty growth scenarios remained recurrent. In the 21st century, dirty development has predominated, with the exception of the 2015–2020 period which, as in Western countries, saw a fall of both emissions and wellbeing due to the pandemic (again, not observed when considering the conventional HDI, see Figure SM4). Lastly, the Middle East has been historically characterised by dirty development and is the world region where decoupling has been the rarest.

In sum, the world and most of its regions experienced a prevalence of dirty development in the earliest decades under study, with GHGe outpacing AHDI. This gave way to increasingly widespread episodes of relative decoupling, as wellbeing expanded faster than emissions. The ideal absolute decoupling scenario, where wellbeing improved and emissions fell in absolute terms, was rare and reversible. Importantly, absolute wellbeing decoupling was not limited to Western countries; rather, it was recurrent in many developing countries. Indeed, some Western countries have recently experienced underdevelopment with emission reduction, as emission reductions are associated not with wellbeing improvements but rather with worsening AHDI levels.

4.3. Modes of decoupling

The decoupling between wellbeing and emissions (when AHDI grows faster than GHGe) can be due to wellbeing outpacing economic growth, to economic growth outpacing emissions, or to a combination of both. In this section we examine which mode of decoupling has predominated in different periods at the global and regional levels.

The transition from one kind of decoupling to another is far from linear, but some global patterns can be discerned from the data. In the late-19th and early-20th centuries, the predominant forms of decoupling were relative impact decoupling (GDP outpacing the increase in GHGe) and wellbeing decoupling (AHDI growing faster than GDP) (Figure 6). Since the 1950s, the leading mode of decoupling was relative impact decoupling, leading in the closing two decades of the century to a period of relative dual decoupling: wellbeing increased faster than incomes, and economic growth outpaced emissions, even if emissions increased in absolute terms. Moreover, decoupling took place with historically leading rates of growth of AHDI, GDP per capita, and GHGe (Figures SM7-10). Globally, the only period of absolute dual decoupling, when emissions actually decreased in a context of expanding economies and rising wellbeing, took place in the aftermath of World War II. This is also the case if we measure wellbeing with the conventional HDI rather than with the AHDI (Figure SM5).

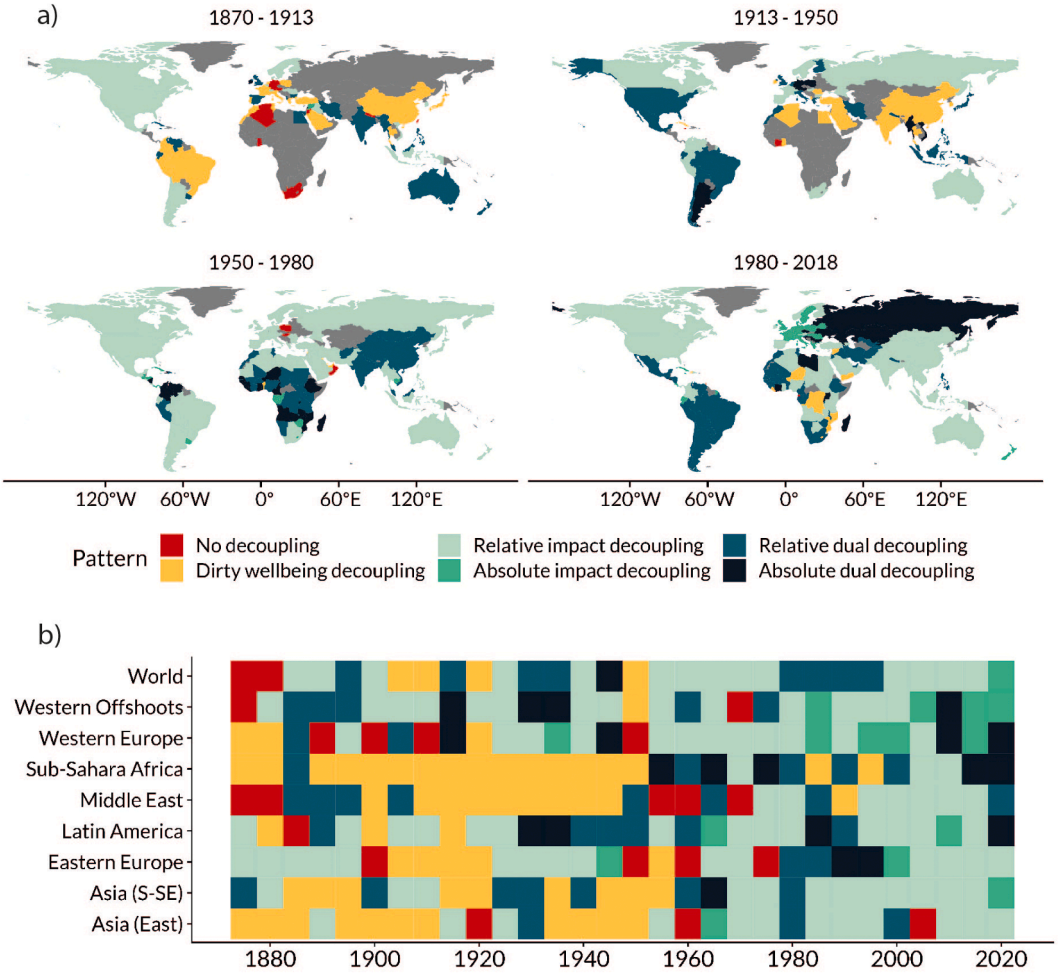


Figure 6. National (a) and regional patterns of combined well-being and decoupling patterns identified in Figure 2b and Table 1. Figure SM5 replicates this exercise using the conventional HDI calculation (without the income component) instead of the AHDI.

These global trends hold much regional variation (Figure 6). In Western countries until the mid-20th century the most common pattern was dirty wellbeing decoupling, occasionally accompanied by impact decoupling (both relative and absolute). In the earlier periods of economic growth under study, AHDI grew faster than GDP per capita in the West. This kind of decoupling has been comparatively rare ever since: incomes have outpaced AHDI since the 1950s, with the exceptions of the periods 2000–2005 and 2010–2015 in Western Europe. Relative impact decoupling became, instead, the most common mode of decoupling in Western societies: economic growth has, as a rule, outpaced both wellbeing indicators and GHGe; the latter have occasionally fallen in absolute terms in a few European countries.

In the rest of the world, we find that dirty wellbeing decoupling predominated until well into the 20th century, both in AHDI as in the conventional HDI (Figure SM5). In Latin America this was often accompanied by impact decoupling (GDP outpacing GHGe) until the 1950s, with several short periods of absolute fall in emissions. Sub-Saharan Africa also experienced dual decoupling, especially since the mid-20th century. These dynamics were especially notable between 1950–1980 and over the last decade, when wellbeing expanded faster than economic growth and total greenhouse gas emissions were reduced. Substantial reductions of GHGe in sub-Saharan Africa were fundamentally due to the slowing down of deforestation processes, rather than to technological improvements, as shown by the fact that fossil-fuel emissions were not reduced (Figure SM3). Eastern Europe shows evidence of dual decoupling during and since the dissolution of the USSR. In these times of crisis, AHDI outpaced economic growth, and economic growth outpaced emissions (Figure SM7-10).

In Asia and the Middle East, we find a lengthy period dominated by a dirty development pattern with emissions and wellbeing expanding faster than the economy, which was followed since c.1970 by a predominance of relative impact decoupling. In very broad terms, this implies that development in these regions used to be efficient from a wellbeing standpoint (non-income components of AHDI improving faster than incomes) but environmentally inefficient (emissions outpacing output), and since c.1970 has been more environmentally efficient (emissions growing more slowly than GDP) but less socially efficient (as GDP per capita growth has not translated into equivalent gains in the non-income components of AHDI).

In sum, notwithstanding large differences between regions and countries, we find that the earlier periods of economic growth were environmentally dirty but socially efficient. Rich countries transitioned since the early or mid-20th century to a scenario of relative impact decoupling, where improvements to wellbeing lagged behind economic growth. In the rest of the world, wellbeing decoupling continued until well into the 20th century, and is still recurrent in many countries today. The joint presence of both forms of decoupling (wellbeing outpacing incomes and economic growth outpacing emissions) has been rare and reversible, both in Western and non-Western countries.

These regional trends suggest a relationship between modes of decoupling and relative levels of economic development. As shown in Figure 7, at lower levels of per capita income, the most common pattern is wellbeing decoupling followed by dual decoupling. That is, in lower income countries (and thus at earlier stages of economic development) it is more common to achieve a scenario where wellbeing improves faster than GDP per capita, and the economy expands faster than emissions. As countries become richer, the combination of both modes of decoupling becomes rarer and impact decoupling (GDP outpacing GHGe) occurs almost exclusively without wellbeing decoupling. This kind of impact decoupling in rich countries is normally 'relative', as GHGe continue to grow even if more slowly than GDP, although at very high-income levels absolute impact decoupling becomes more common, with GDP rising and emissions falling.

Therefore, the aggregate evidence of national experiences implies that in rich countries economic growth is socially less efficient but environmentally more efficient. The data also suggest that emission mitigation, occurring as it does mostly in high-income economies, takes place only at sufficiently high levels of per capita emissions. The level of environmental impacts at which decoupling occurs is crucial to the prospects of moving towards a 'safe and just operating space' for economic and social development. Let us look more closely at the levels of wellbeing and emissions at which decoupling has historically taken place.

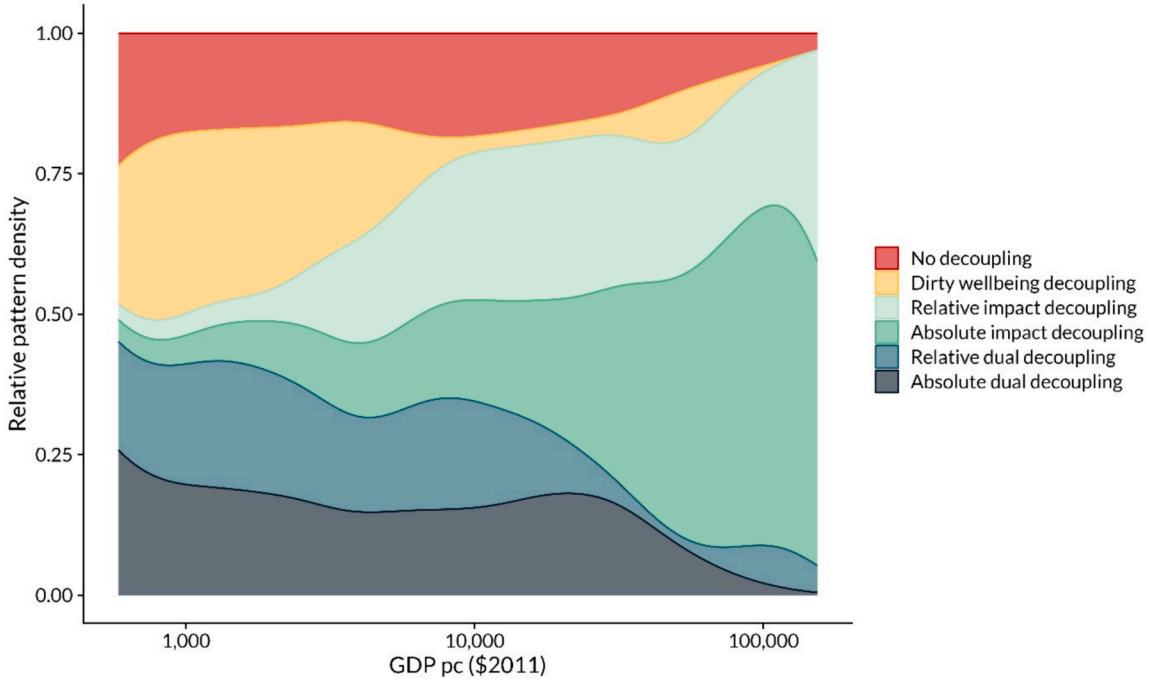


Figure 7. Relative density of country-year cases of dual decoupling patterns (as defined in Figure 2b). Figure SM6 replicates this exercise using the conventional HDI calculation (without the income component) instead of the AHDI.

4.4. The levels of decoupling

Looking at decoupling is useful inasmuch as it reveals the compared trajectories of wellbeing and GHGe. However, decoupling is only an indicator of how to move towards a goal: achieving high levels of wellbeing within a climate safety space. In other words, finding a trajectory leading to a 'safe and just operating space' for human societies. How close have historical decoupling trajectories come to this goal?

Figure 8a and 8b shows the correlation between levels of AHDI and emissions per capita at a regional and national level, highlighting a selection of countries. The average global emissions over the last 150 years have been of 5.9 tons of CO_2e per capita, while the average value of AHDI has been of 0.21. During most of the period under study, Western countries have been persistently above the global average of both variables. At first, the rest of the world released emissions below the global average, although many developing countries went on to exceed it. Convergence in terms of AHDI has been slower and more recent. Sub-Saharan Africa is still below the global historical average of the non-income components of the wellbeing indicators.

Regional and national trajectories reveal that the relationship between AHDI and GHGe is positive but not linear. Some countries, such as China, have increased their environmental impact far faster than their wellbeing levels. Western countries have seen their AHDI grow faster than their emissions, although their emission levels were already much higher than the rest of the world by 1870. What all world regions and countries do share, however, is that no development model has managed to produce high human development indicators with low environmental impact. With extremely few exceptions (India since the late-20th century), no country has released fewer per capita emissions than the global historical average while at the same time achieving higher human development levels than the global mean.

Moreover, all historical trajectories are very far from what we would today consider a 'safe and just operating space'. The precise boundaries of such a space (a lower-bound level of wellbeing and an upper-bound

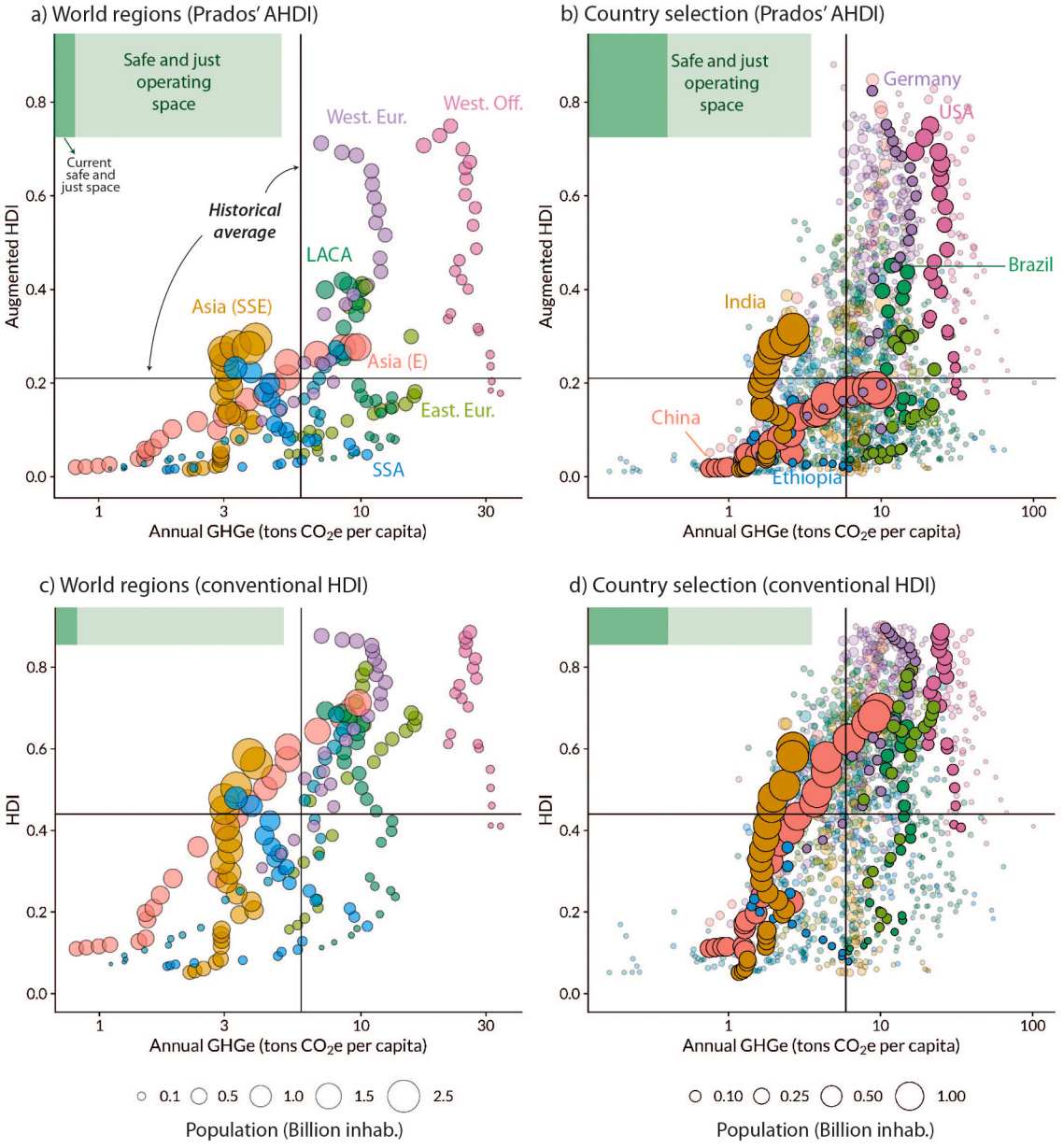


Figure 8. Relation between Augmented HDI (y-axis) and per capita GHGe in tons of CO_2e in world regions (a) and world countries (b); and between United Nations' HDI and per capita GHGe in tons of CO_2e in world regions (c) and world countries (d). The green box represents the historical safe and just operating space. The dark green part of the box represents the current safe and just operating space.

level of environmental impact) are difficult to define. As detailed in the Methods section above, we assume that a just operating space would involve levels of wellbeing approaching the average enjoyed by Western countries since 2000 (an AHDI of ~ 0.7 or higher) and a level of per capita emissions compatible with the Paris Climate Agreement's goal of keeping global emissions from rising more than 2 °C above pre-industrial levels by 2050 (3.5 t CO_2e hab^{-1} historically and 0.4 CO_2e hab^{-1} at present). With these thresholds, depicted in Figure 8, no country or world region has ever come close to a safe and just scenario. Some regions and countries (such as Western Europe) are moving in that direction, but their levels are still very far away from a sustainable operating space thus defined.

However, this image becomes more nuanced if we consider a conventional, United Nations-stlye HDI instead of Prados's AHDI (Figure 8c and 8d). As discussed earlier, the key differences between these two measures of wellbeing is that the latter transforms the educational attainment and life expectancy values to emphasise improvements at higher levels, while also inlcuding a 'liberal democracy index' as a proxy for political freedom. When using the conventional HDI we also find no countries in the 'safe and just operating space.' Likewise, very few countries find themselves above the historical wellbeing averages (0.41 for the HDI) while also being below historical average emission levels. One exception is reform-era China between 1980 and 2010. While Western countries almost do not change position in the graph relative to the AHDI version, the picture of the rest of the world is substantially improved by taking the conventional HDI as a measure of wellbeing. In general, developing countries have achieved substantial improvements to wellbeing (as measured by the HDI) with fewer emissions per capita than Western countries before them. Nevertheless, even with this metric, all countries seem to be moving in the opposite direction to the 'safe and just operating space.'.

5. Discussion

5.1. How the world became rich… and dirty

The Industrial Revolution ushered in a world of unprecedently sustained economic growth: global per capita GDP multiplied by a factor of 20 in the last 200 years (Bolt and van Zanden, 2020). The results in terms of human wellbeing were formidable, with life expectancy, access to education, and other key social indicators increasing dramatically (Deaton, 2013; van Zanden et al., 2021a). And yet, the correlation between income and wellbeing in the long run is far from perfect, given that human wellbeing, regardless of which composite indicator we choose, is multidimensional and thus depends on factors which are not solely driven by a rise in output per head (Prados de la Escosura, 2022a; van Zanden et al., 2021a; van Zanden et al., 2021b). As we showed in the previous sections, wellbeing tended to improve faster in the earlier stages of economic growth. As has long been suggested by scholars (Max-Neef, 1995), in later periods GDP growth does not lead to proportional improvements in AHDI. Western countries were very efficient at turning GDP growth into faster wellbeing improvements until the early-20th century, while in recent decades this only occurs in lower income countries.

The Industrial Revolution and, especially, the global spread of industrialization also led to substantially higher per capita environmental footprints than ever before. As shown extensively by E.A. Wrigley (1990, 2013, 2016), all pre-industrial economies were 'organic' in the sense that they depended ultimately on photosynthesis to fuel their production processes. Steam engines unlocked the potential of fossil fuels, which became 'subterranean forests' (Sieferle, 2001) and opened up a new 'vertical frontier' (Barbier, 2010). This allowed Britain first, then north-western Europe, and eventually much of the world to escape the land constraints that had long limited economic growth. Fossil fuels were not optional for modern economic development, and their increasing use reshaped humans' relationship to their environment. These changes reached the countryside too: industrial demand for land-based raw materials and the food requirements of rising populations led to a far-reaching expansion of the agricultural frontier (Barbier, 2010). Indeed, global land-based greenhouse gas emissions were larger than fossil-fuel emissions until well into the 20th century.

In this paper we have examined the evidence of a joint process of wellbeing improvement and planetary health degradation. Rising standards of living have implied larger environmental impacts in all world regions. Modern economic growth has led, without exceptions, to unprecedented levels of wellbeing while also transgressing planetary boundaries. Yet, not all countries have equally efficient in their ability to sustain wellbeing improvements relative to their environmental impact. As shown above, non-Western countries have in general achieved levels of wellbeing at lower emission levels than Western countries had managed in the past. This could be due to the fact that rich countries achieved higher wellbeing levels at a time when technology was less environmentally efficient. But it is also because some developing countries have been capable of producing more wellbeing per unit of economic output. There is a large literature that suggests that some developing countries are able to achieve high levels in key wellbeing indicators (such as life expectancy) with comparatively modest emission levels (Roberts et al., 2020; Steinberger & Roberts, 2010). However, in a long-term perspective and considering composite indicators like the HDI, our results show that all countries with wellbeing levels substantially higher than the global historical average are also responsible for levels of emissions far beyond the mean. As countries became richer, they released more emissions per capita. Even if this process has slowly been reversed at very high-income levels in recent decades (Le Quéré et al., 2019; Vogel & Hickel, 2023) and developing countries are increasingly efficient at improving human wellbeing relative to emissions, no country has historically been remotely near a 'safe and just operating space'.

5.2. Historical foundations for decoupling strategies

Decoupling material wellbeing from environmental degradation, especially anthropogenic climate change, is a central goal for world development in the 21st century (OECD, 2011; UNEP, 2011b; World Bank, 2011). The mainstream development literature considers absolute impact decoupling (increasing GDP while decreasing emissions) a plausible strategy to achieve that goal. For instance, the shared socioeconomic pathways framework used by the IPCC consider different mitigation scenarios under sustained economic growth (Riahi et al., 2017; van Vuuren et al., 2017). Nevertheless, there is a substantial stream of literature which is sceptic about the possibility of maintaining current rates of economic growth while mitigating emissions. The idea that efficiency gains might not result in actual environmental savings but rather in an absolute increase in resource use (and thus emissions) is of course an old one, harking back to Jevons' paradox (Alcott, 2005). But many scholars have further argued that even if absolute impact decoupling is indeed taking place in some high-income countries, it has shown itself to be reversible (Hubacek et al., 2021) and the emission reductions achieved are insufficient to achieve international climate agreements (Hickel & Kallis, 2020; Parrique et al., 2019). According to our study, the world has transitioned from an era of recurrent episodes of dirty development (emissions outpacing wellbeing improvements) to a period of widespread relative impact decoupling, with recent cases of absolute impact decoupling in some rich Western countries. However, we also find that absolute decoupling is not new nor exclusively western: Latin America and sub-Saharan Africa experienced absolute impact decoupling in the past, especially in contexts of a reduction of land-based emissions.

Faced with the uncertainty of whether impact decoupling will be fast enough to avoid the worsening of climate change, some argue for a focus on decoupling wellbeing from economic growth in order to stay within

an environmentally safe operating space (Jackson, 2021; Raworth, 2017; Roberts et al., 2020). Our results suggest that this kind of decoupling was generalised in the past while today it only occurs in lower income countries. In the West wellbeing only outpaced GDP in the late-19th and early-20th centuries, encouraged by the rapid increases in life expectancy that came with the earlier periods of modern economic growth (Prados de la Escosura, 2022a). This kind of decoupling also occurred in interwar Europe, which might be surprising: these were after all decades dominated by political turmoil, violence, and economic downturns. Nevertheless, mortality fell substantially across Europe, and a shift from agriculture to industry led to technological innovation and improvements in literacy and education levels (Millward & Baten, 2010). Wellbeing indicators are, after all, 'in part stock rather than flow' (Prados de la Escosura, 2022b), and are thus less susceptible to short-term changes than GDP growth in the face of political and economic crises. Furthermore, synthetic indicators of wellbeing like the AHDI always exclude fundamental dimensions of human welfare, such as the suffering caused by wars and their aftermath (Judt, 2006). In the rest of the world, wellbeing mostly kept improving faster than average incomes until well into the 20th century and, in some regions, continues to do so. This can be partially explained because these regions have successfully adapted and adopted innovations in sanitation and healthcare which have increased life expectancy at a faster pace than GDP per capita growth. A similarly remarkable expansion in public education, particularly at the primary level, contributes to explain the comparatively lower environmental cost of human wellbeing improvements in sub-Saharan Africa and Asia. These improvements are even more impressive if we consider the scale of population growth in these regions over the last 100 years.

In sum, the historical evidence suggests that wellbeing decoupling is less frequent as countries become richer, an issue with important implications for the specialist literature in economics (Kahneman & Deaton, 2010). At first sight, this would suggest that there is a limited scope for decoupling wellbeing from GDP in the future, as more countries reach higher income levels. However, this could instead suggest that limiting economic growth in high income countries could, given the right context, have a negligible negative effect on people's wellbeing (Roberts et al., 2020).

5.3. A historical 'safe and just operating space'

Decoupling is not a goal in itself but a means to an end: achieving or maintaining high levels of wellbeing while mitigating environmental impacts. In fact, evidence of decoupling between rates of growth of AHDI and GHGe can be misleading, as it could occur in contexts of decreasing wellbeing, or hide an increase of environmental impacts in absolute terms.

Recent studies have measured whether countries have surpassed 'planetary boundaries' and whether they have been capable of satisfying people's needs, defined as the 'finite number of basic human needs that are universal, satiable and non-substitutable' (O'Neill et al., 2018). This literature has revealed that, although some countries have achieved high levels of wellbeing with comparatively low environmental impacts (Steinberger & Roberts, 2010), no country is currently within a 'safe and just operating space' (Fanning et al., 2022; O'Neill et al., 2018). We cast our nets further back into the past and find that at no point in modern history has any country reached that safe and just operating space. As economies developed, their trajectories took them around the historical safe and just operating space: from the bottom-left quadrant of not enough wellbeing to the top-right quadrant of too much environmental impact (Figure 8). While modern economic growth has led in all world regions to unprecedented improvements in terms of human wellbeing, expanding life expectancy, education, and political freedom (Deaton, 2013; Prados de la Escosura, 2022a; van Zanden et al., 2021a), at the same time this progress has brought about increases in per capita emissions that place human societies outside planetary boundaries (Dearing et al., 2014; Steffen et al., 2015). This relationship is not linear nor do all world countries share the same patterns, because of structural differences and because their development processes took place in different historical contexts. And yet, looking at the past it is difficult to ignore the fact that the trajectory of all world regions and countries seems to move in the opposite direction to the achievement of a good life for all within planetary boundaries.

Modern economic development has made the world rich and as a result people live, on average, longer, and are better educated and freer than ever before. But it has operated, persistently and everywhere, outside environmental boundaries. If sustainable development implies finding a place within a safe and just operating space, then a different pathway is necessary: one capable of dissociating wellbeing from economic growth and economic growth from environmental impacts. Recent studies argue that achieving this twin aims is possible (Schlesier et al., 2024). Our own work, meanwhile, suggests that to do so substantial changes are required, both in accelerating the efficiency gains of technological change and in designing and applying social policies to expand wellbeing more effectively.

It is important to note that our indicators are limited to rough proxies of wellbeing and environmental impacts, measured as levels of HDI and GHGe. The HDI indicator misses important components of wellbeing such as social support, equality, and nutritional standards. GHGe as a proxy of environmental impacts has two main problems. One is that it misses the multiple dimensions of planetary degradation, which include at least 9 planetary boundaries. The other is related to the implicit uncertainty of the historical assessment of the climate planetary boundary, mainly because it involves making arbitrary assumptions on the allocation of the GHGe budget to people living in different countries and epochs.

6. Conclusion

Improving people's wellbeing while preventing further disruption of the Earth systems can only be achieved by decoupling human development from its environmental impacts. A major scholarly and political debate surrounds decoupling strategies: green growth supporters defend plans to increase GDP per capita without further expanding emissions and resource extraction; post-growth advocates propose to de-link wellbeing from the imperative of economic growth, and thus also from its ecological consequences. In this paper, we examine global and regional trends of wellbeing, income, and greenhouse emissions over the last 150 years. Using a simple framework, we identify periods and places with incidence of growth-impact decoupling (the kind championed by green growth proposals) and wellbeing-income decoupling (as advocated by post-growth theories). We find that both forms of decoupling have historical precedents, even if these are unevenly distributed: impact decoupling was more common and larger in richer countries, wellbeing decoupling more widespread in poorer regions and at lower income levels. Across regions, history suggests that when both forms of decoupling were present at the same time, more substantial mitigation was achieved – a lesson that matters greatly for our present and future. History also holds a more sobering lesson: our results imply that no country has ever shown a development model compatible with what we today identify as a 'safe and just operating space'.

Modern economic growth is the hero in the story economic historians usually tell, having freed societies from ancient and long-standing shackles which limited output per head and made mass poverty inevitable. But it can also become the ogre of the tale, bringing about further climate change to threaten prosperity. Will developing regions converge faster in terms of wellbeing than in their per capita environmental impact? Will richer countries continue their historical trends centred on impact decoupling or also manage to decouple wellbeing from income growth? We do not have the answers, among other things because they depend on the outcomes of decoupling debates, for which history is a partial but irreplaceable guide.

CRediT authorship contribution statement

Juan Infante-Amate: Writing – original draft, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Emiliano Travieso:** Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis, Conceptualization. **Eduardo Aguilera:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

All authors disclose any financial and personal relationships with other people or organizations that could inappropriately influence (bias) their work.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.worlddev.2024.106754.

References

Akizu-Gardoki, O., Bueno, G., Wiedmann, T., Lopez-Guede, J. M., Arto, I., Hernandez, P., & Moran, D. (2018). Decoupling between human development and energy consumption within footprint accounts. *Journal of Cleaner Production, 202*, 1145–1157.

Alcott, B. (2005). Jevons' paradox. *Ecological Economics, 54*(1), 9–21.

Alkire, S. (2002). Dimensions of human development. *World development, 30*(2), 181–205.

Allen, R. C. (2001). The great divergence in European wages and prices from the Middle Ages to the First World War. *Explorations in Economic History, 38*(4), 411–447. https://doi.org/10.1006/exeh.2001.0775

Allen, R. C. (2011). *Global Economic History: A Very Short Introduction*. Oxford University Press.

Andrew, R. M., & Peters, G. P. (2021). The Global Carbon Project's fossil CO2 emissions dataset. https://doi.org/10.5281/ZENODO.5569235.

Barbier, E. B. (2010). *Scarcity and frontiers: how economies have developed through natural resource exploitation*. Cambridge University Press.

Bithas, K., & Kalimeris, P. (2018). Unmasking decoupling: Redefining the resource intensity of the economy. *Science of the Total Environment, 619*, 338–351.

Bolt, J., & van Zanden, J. L. (2020). Maddison style estimates of the evolution of the world economy. *A new 2020 update. Maddison-Project Working Paper WP-15*.

Brolin, J., & Kander, A. (2022). Global trade in the Anthropocene: A review of trends and direction of environmental factor flows during the Great Acceleration. *The Anthropocene Review, 9*, 71–110. https://doi.org/10.1177/2053019620973711

Chancel, L., & Piketty, T. (2021). Global income inequality, 1820–2020: The persistence and mutation of extreme inequality. *Journal of the European Economic Association, 19*(6), 3025–3062.

Coppedge, M., John Gerring, Carl Henrik Knutsen, Staffan I. Lindberg, Jan Teorell, David Altman, Michael Bernhard, Agnes Cornell, M. Steven Fish, Lisa Gastaldi, Haakon Gjerløw, Adam Glynn, Ana Good God, Sandra Grahn, Allen Hicken, Katrin Kinzelbach, Joshua Krusell, Kyle L. Marquardt, Kelly McMann, Valeriya Mechkova, Juraj Medzihorsky, Natalia Natsika, Anja Neundorf, Pamela Paxton, Daniel Pemstein, Josefine Pernes, Oskar Rydén, Johannes von Römer, Brigitte Seim, Rachel Sigman, Svend-Erik Skaaning, Jeffrey Staton, Aksel Sundström, Eitan Tzelgov, Yi-ting Wang, Tore Wig, Steven Wilson and Daniel Ziblatt. (2023). *V-Dem [Country-Year/Country-Date] Dataset v13" Varieties of Democracy (V-Dem) Project*. https://doi.org/https://doi.org/10.23696/vdemds23.

Costanza, R., Kubiszewski, I., Giovannini, E., Lovins, H., McGlade, J., Pickett, K. E., Ragnarsdóttir, K. V., Roberts, D., De Vogli, R., & Wilkinson, R. (2014). Development: Time to leave GDP behind. *Nature, 505*(7483), 283–285.

Dasgupta, S., Laplante, B., Wang, H., & Wheeler, D. (2002). Confronting the environmental Kuznets curve. *Journal of Economic Perspectives, 16*(1), 147–168. https://doi.org/10.1257/0895330027157

Dearing, J. A., Wang, R., Zhang, K., Dyke, J. G., Haberl, H., Hossain, M. S., Langdon, P. G., Lenton, T. M., Raworth, K., Brown, S., Carstensen, J., Cole, M. J., Cornell, S. E., Dawson, T. P., Doncaster, C. P., Eigenbrod, F., Floerke, M., Jeffers, E., Mackay, A. W., Nykvist, B., & Poppy, G. M. (2014). Safe and just operating spaces for regional social-ecological systems. *Global Environmental Change-Human and Policy Dimensions, 28*, 227–238. https://doi.org/10.1016/j.gloenvcha.2014.06.012

Deaton, A. (2013). *The Great Escape: Health, Wealth, and the Origins of Inequality*. Princeton University Press.

Dinda, S. (2004). Environmental Kuznets curve hypothesis: A survey. *Ecological Economics, 49*(4), 431–455.

Fanning, A. L., O'Neill, D. W., Hickel, J., & Roux, N. (2022). The social shortfall and ecological overshoot of nations. *Nature Sustainability, 5*(1), 26–36. https://doi.org/10.1038/s41893-021-00799-z

Friedlingstein, P., Jones, M. W., O'Sullivan, M., Andrew, R. M., Bakker, D. C. E., Hauck, J., Le Quéré, C., Peters, G. P., Peters, W., Pongratz, J., Sitch, S., Canadell, J. G., Ciais, P., Jackson, R. B., Alin, S. R., Anthoni, P., Bates, N. R., Becker, M., Bellouin, N., Bopp, L., Chau, T. T. T., Chevallier, F., Chini, L. P., Cronin, M., Currie, K. I., Decharme, B., Djeutchouang, L. M., Dou, X., Evans, W., Feely, R. A., Feng, L., Gasser, T., Gilfillan, D., Gkritzalis, T., Grassi, G., Gregor, L., Gruber, N., Gürses, Ö., Harris, I., Houghton, R. A., Hurtt, G. C., Iida, Y., Ilyina, T., Luijkx, I. T., Jain, A., Jones, S. D., Kato, E., Kennedy, D., Klein Goldewijk, K., Knauer, J., Korsbakken, J. I., Körtzinger, A., Landschützer, P., Lauvset, S. K., Lefèvre, N., Lienert, S., Liu, J., Marland, G., McGuire, P. C., Melton, J. R., Munro, D. R., Nabel, J. E. M. S., Nakaoka, S. I., Niwa, Y., Ono, T., Pierrot, D., Poulter, B., Rehder, G., Resplandy, L., Robertson, E., Rödenbeck, C., Rosan, T. M., Schwinger, J., Schwingshackl, C., Séférian, R., Sutton, A. J., Sweeney, C., Tanhua, T., Tans, P. P., Tian, H., Tilbrook, B., Tubiello, F., van der Werf, G. R., Vuichard, N., Wada, C., Wanninkhof, R., Watson, A. J., Willis, D., Wiltshire, A. J., Yuan, W., Yue, C., Yue, X., Zaehle, S., & Zeng, J. (2022). Global Carbon Budget 2021. *Earth Syst. Sci. Data, 14*(4), 1917–2005. https://doi.org/10.5194/essd-14-1917-2022

Gales, B., Kander, A., Malanima, P., & Rubio, M. (2007). North versus South: Energy transition and energy intensity in Europe over 200 years. *European Review of Economic History, 11*(2), 219–253.

Gardner, L., & Roy, T. (2020). *The Economic History of Colonialism*. Bristol University Press.

Gasser, T., Crepin, L., Quilcaille, Y., Houghton, R. A., Ciais, P., & Obersteiner, M. (2020). Historical CO2 emissions from land use and land cover change and their uncertainty. *Biogeosciences, 17*(15), 4075–4101. https://doi.org/10.5194/bg-17-4075-2020

Gierlinger, S., & Krausmann, F. (2012). The physical economy of the United States of America: Extraction, trade, and consumption of materials from 1870 to 2005. *Journal of Industrial Ecology, 16*(3), 365–377.

Gütschow, J., Jeffery, M. L., Gieseke, R., Gebel, R., Stevens, D., Krapp, M., & Rocha, M. (2016). The PRIMAP-hist national historical emissions time series. *Earth System Science Data, 8*(2), 571–603. https://doi.org/10.5194/essd-8-571-2016

Gütschow, J., & Pflüger, M. (2023). *The PRIMAP-hist national historical emissions time series v2.4.2 (1750-2021)*. zenodo. doi:10.5281/zenodo.7727475.

Haas, W., Krausmann, F., Wiedenhofer, D., Lauk, C., & Mayer, A. (2020). Spaceship earth's odyssey to a circular economy-a century long perspective. *Resources, Conservation and Recycling, 163*, Article 105076.

Haberl, H., Wiedenhofer, D., Virág, D., Kalt, G., Plank, B., Brockway, P., Fishman, T., Hausknost, D., Krausmann, F., & Leon-Gruchalski, B. (2020). A systematic review of the evidence on decoupling of GDP, resource use and GHG emissions, part II: Synthesizing the insights. *Environmental Research Letters, 15*(6), Article 065003.

Hansis, E., Davis, S. J., & Pongratz, J. (2015). Relevance of methodological choices for accounting of land use change carbon fluxes. *Global Biogeochemical Cycles, 29*(8), 1230–1246. https://doi.org/10.1002/2014GB004997

Henriques, S. T., & Borowiecki, K. J. (2017). The drivers of long-run CO2 emissions in Europe, North America and Japan since 1800. *Energy Policy, 101*, 537–549. https://doi.org/10.1016/j.enpol.2016.11.005

Hickel, J. (2017). *The divide: A brief guide to global inequality and its solutions*. Random House.

Hickel, J., & Kallis, G. (2020). Is Green Growth Possible? *New Political Economy, 25*(4), 469–486. https://doi.org/10.1080/13563467.2019.1598964

Hickel, J., Kallis, G., Jackson, T., O'Neill, D. W., Schor, J. B., Steinberger, J. K., Victor, P. A., & Ürge-Vorsatz, D. (2022). Degrowth can work—here's how science can help. *Nature, 612*(7940), 400–403.

Hoekstra, R. (2019). *Replacing GDP by 2030: Towards a common language for the well-being and sustainability community*. Cambridge University Press.

Hossain, M. A., & Chen, S. (2021). Nexus between Human Development Index (HDI) and CO 2 emissions in a developing country: Decoupling study evidence from Bangladesh. *Environmental Science and Pollution Research, 28*, 58742–58754.

Houghton, R. A., & Nassikas, A. A. (2017). Global and regional fluxes of carbon from land use and land cover change 1850–2015. *Global Biogeochemical Cycles, 31*(3), 456–472. https://doi.org/10.1002/2016GB005546

Hubacek, K., Chen, X., Feng, K., Wiedmann, T., & Shan, Y. (2021). Evidence of decoupling consumption-based CO2 emissions from economic growth. *Advances in Applied Energy, 4*, Article 100074.
Infante-Amate, J., Soto, D., Aguilera, E., García-Ruiz, R., Guzmán, G., Cid, A., & González de Molina, M. (2015). The Spanish transition to industrial metabolism: Long-term material flow analysis (1860–2010). *Journal of Industrial Ecology, 19*(5), 866–876.
IPCC. (2022). *Mitigation of Climate Change. Working Group III Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (A. K. Shukla, J. Skea, A. Reisinger, R. Slade, R. Fradera, M. Pathak, A. A. Khourdajie, M. Belkacemi, R. van Diemen, A. Hasija, G. Lisboa, S. M. Luz, J. Malley, D. McCollum, S. Some, & P. Vyas, Eds.). Cambridge University Press.
Jackson, T. (2021). *Post growth: Life after capitalism*. John Wiley & Sons.
Jackson, T., & Victor, P. A. (2019). Unraveling the claims for (and against) green growth. *Science, 366*(6468), 950. https://doi.org/10.1126/science.aay0749
Jones, M. W., Peters, G. P., Gasser, T., Andrew, R. M., Schwingshackl, C., Gütschow, J., Houghton, R. A., Friedlingstein, P., Pongratz, J., & Le Quéré, C. (2023). National contributions to climate change due to historical emissions of carbon dioxide, methane, and nitrous oxide since 1850. *Scientific Data, 10*(1), 155.
Judt, T. (2006). *Postwar: A history of Europe since 1945*. Penguin.
Kahneman, D., & Deaton, A. (2010). High income improves evaluation of life but not emotional well-being. *Proceedings of the National Academy of Sciences, 107*(38), 16489–16493.
Kalimeris, P., Bithas, K., Richardson, C., & Nijkamp, P. (2020). Hidden linkages between resources and economy: A "Beyond-GDP" approach using alternative welfare indicators. *Ecological Economics, 169*, Article 106508.
Kallis, G., Kerschner, C., & Martinez-Alier, J. (2012). The economics of degrowth. *Ecological Economics, 84*, 172–180. https://doi.org/10.1016/j.ecolecon.2012.08.017
Kander, A., Malanima, P., & Warde, P. (2014). *Power to the People. Energy in Europe over the Last Five Centuries*. Princeton University Press.
Koyama, M., & Rubin, J. (2022). *How the world became rich: The historical origins of economic growth*. John Wiley & Sons.
Krausmann, F., Gaugl, B., West, J., & Schandl, H. (2016). The metabolic transition of a planned economy: Material flows in the USSR and the Russian Federation 1900 to 2010. *Ecological Economics, 124*, 76–85.
Krausmann, F., Gingrich, S., Eisenmenger, N., Erb, K.-H., Haberl, H., & Fischer-Kowalski, M. (2009). Growth in global materials use, GDP and population during the 20th century. *Ecological Economics, 68*(10), 2696–2705.
Krausmann, F., Schandl, H., & Sieferle, R. P. (2008). Socio-ecological regime transitions in Austria and the United Kingdom. *Ecological Economics, 65*(1), 187–201.
Kuznets, S. (1955). Economic Growth and Income Inequality. *The American Economic Review, 45*(1), 1–28.
Le Quéré, C., Korsbakken, J. I., Wilson, C., Tosun, J., Andrew, R., Andres, R. J., Canadell, J. G., Jordan, A., Peters, G. P., & van Vuuren, D. P. (2019). Drivers of declining CO2 emissions in 18 developed economies. *Nature Climate Change, 9*(3), 213–217.
Li, D., Shen, T., Wei, X., & Li, J. (2022). Decomposition and decoupling analysis between HDI and carbon emissions. *Atmosphere, 13*(4), 584.
Lindmark, M. (2004). Patterns of historical CO2 intensity transitions among high and low-income countries. *Explorations in Economic History, 41*(4), 426–447.
Magalhães, N., Fressoz, J.-B., Jarrige, F., Le Roux, T., Levillain, G., Lyautey, M., Noblet, G., & Bonneuil, C. (2019). The physical economy of France (1830–2015). The history of a parasite? *Ecological Economics, 157*, 291–300.
McNeill, J. R., & Engelke, P. (2014). *The great acceleration: An environmental history of the anthropocene since 1945*. The Belknap Press of Harvard University Press.
Milanovic, B. (2013). Global income inequality in numbers: In history and now. *Global policy, 4*(2), 198–208.
Millward, R., & Baten, J. (2010). Population and living standards, 1914-1945. In S. Broadberry & K. H. O'Rourke (Eds.), *The Cambridge Economic History of Modern Europe. Volume 2: 1870 to the Present* (pp. 233-263). Cambridge University Press.
O'Neill, D. W., Fanning, A. L., Lamb, W. F., & Steinberger, J. K. (2018). A good life for all within planetary boundaries. *Nature Sustainability, 1*(2), 88–95. https://doi.org/10.1038/s41893-018-0021-4
OECD. (2011). *Towards green growth*. Organization for Economic Co-Operation and Development.
OECD. (2023). *Real GDP long-term forecast*. https://data.oecd.org/gdp/real-gdp-long-term-forecast.htm.
Panayotou, T. (1993). Empirical tests and policy analysis of environmental degradation at different stages of economic development.
Parrique, T., Barth, J., Briens, F., Kerschner, C., Kraus-Polk, A., Kuokkanen, A., & Spangenberg, J. H. (2019). Decoupling debunked. *Evidence and arguments against green growth as a sole strategy for*. sustainability.
Pearlstine, E. V. (2022). *Scent: A natural history of fragrance*. Yale University Press.
Peters, G. P., & Hertwich, E. G. (2008). CO2 embodied in international trade with implications for global climate policy. *Environmental Science & Technology, 42*(5), 1401–1407. https://doi.org/10.1021/es072023k
Pomeranz, K. (2000). *The Great Divergence*. China, Europe and the Making of the Modern World Economy: Princeton University Press.
Prados de la Escosura, L. (2022a). *Human Development and the Path to Freedom. 1870 to the present*. Cambridge University Press.
Prados de la Escosura, L. (2022b). A not-so-silly debate! Response to Jan Luiten van Zanden's review. *Rivista di storia economica, 38*(3), 1–5.
Raworth, K. (2017). *Doughnut economics: Seven ways to think like a 21st-century economist*. Chelsea Green Publishing.
Riahi, K., Van Vuuren, D. P., Kriegler, E., Edmonds, J., O'neill, B. C., Fujimori, S., Bauer, N., Calvin, K., Dellink, R., & Fricko, O. (2017). The Shared Socioeconomic Pathways and their energy, land use, and greenhouse gas emissions implications: An overview. *Global environmental change, 42*, 153-168.
Roberts, J. T., Steinberger, J. K., Dietz, T., Lamb, W. F., York, R., Jorgenson, A. K., Givens, J. E., Baer, P., & Schor, J. B. (2020). Four agendas for research and policy on emissions mitigation and well-being. *Global Sustainability, 3*, e3.
Rockström, J., Gupta, J., Qin, D., Lade, S. J., Abrams, J. F., Andersen, L. S., Armstrong McKay, D. I., Bai, X., Bala, G., & Bunn, S. E. (2023). Safe and just Earth system boundaries. *Nature*, 1–10.
Rockstrom, J., Steffen, W., Noone, K., Persson, A., Chapin, F. S., III, Lambin, E. F., Lenton, T. M., Scheffer, M., Folke, C., Schellnhuber, H. J., Nykvist, B., de Wit, C. A., Hughes, T., van der Leeuw, S., Rodhe, H., Sorlin, S., Snyder, P. K., Costanza, R., Svedin, U., Falkenmark, M., Karlberg, L., Corell, R. W., Fabry, V. J., Hansen, J., Walker, B., Liverman, D., Richardson, K., Crutzen, P., & Foley, J. A. (2009). A safe operating space for humanity. *Nature, 461*(7263), 472–475. https://doi.org/10.1038/461472a
Savage, M. (2021). *The return of inequality : Social change and the weight of the past*. Harvard University Press.
Schlesier, H., Schäfer, M., & Desing, H. (2024). Measuring the Doughnut: A good life for all is possible within planetary boundaries. *Journal of Cleaner Production, 141447*.
Shuai, C., Chen, X., Wu, Y., Zhang, Y., & Tan, Y. (2019). A three-step strategy for decoupling economic growth from carbon emission: Empirical evidences from 133 countries. *Science of the Total Environment, 646*, 524–543.
Sieferle, R. P. (2001). *The subterranean forest: Energy systems and the industrial revolution*. White Horse Press.
Steffen, W., Broadgate, W., Deutsch, L., Gaffney, O., & Ludwig, C. (2015). The trajectory of the Anthropocene: The Great Acceleration. *Anthropocene Review, 2*(1), 81–98. https://doi.org/10.1177/2053019614564785
Steinberger, J. K., & Roberts, J. T. (2010). From constraint to sufficiency: The decoupling of energy and carbon from human needs, 1975–2005. *Ecological Economics, 70*(2), 425–433.
Stern, D. I. (2004). The rise and fall of the environmental Kuznets curve. *World Development, 32*(8), 1419–1439. https://doi.org/10.1016/j.worlddev.2004.03.004
Stern, N., & Stiglitz, J. E. (2023). Climate change and growth. *Industrial and Corporate Change, 32*(2), 277–303.
UNEP. (2011a). *Decoupling natural resource use and environmental impacts from economic growth*. United Nations Environment Programme/Earthprint.
UNEP. (2011b). *Towards a green economy: Pathways to sustainable development and poverty eradication*. United Nations Environment Programme.
United Nations. (2022). *World Pouplation Prospects 2022*. https://population.un.org/wpp/Download/Standard/MostUsed/.
Van den Bergh, J. C. (2011). Environment versus growth—A criticism of "degrowth" and a plea for "a-growth". *Ecological Economics, 70*(5), 881–890.
van Vuuren, D. P., Stehfest, E., Gernaat, D. E. H. J., Doelman, J. C., Van den Berg, M., Harmsen, M., de Boer, H. S., Bouwman, L. F., Daioglou, V., Edelenbosch, O. Y., Girod, B., Kram, T., Lassaletta, L., Lucas, P. L., van Meijl, H., Mueller, C., van Ruijven, B. J., van der Sluis, S., & Tabeau, A. (2017). Energy, land-use and greenhouse gas emissions trajectories under a green growth paradigm. *Global Environmental Change-Human and Policy Dimensions, 42*, 237–250. https://doi.org/10.1016/j.gloenvcha.2016.05.008
van Zanden, J. L., Rijpma, A., Malinowski, M., & Mira d'Ercole, M. (2021). *How was life? Volume II: New perspectives on well-being and global inequality since 1820*. OECD.
van Zanden, J. L., van Leeuwen, B., & Xu, Y. (2021). Consequences of growth: Living standards and inequality. In *The Cambridge Economic History of the Modern World, Volume 1 (1700-1870)* (pp. 391-411). Cambridge University Press.
Vogel, J., & Hickel, J. (2023). Is green growth happening? An empirical analysis of achieved versus Paris-compliant CO2–GDP decoupling in high-income countries. *The Lancet Planetary Health, 7*(9), e759–e769.
Wiedmann, T. (2016). Impacts embodied in global trade flows. *Taking stock of industrial ecology*, 159–180.
Wiedmann, T., Lenzen, M., Keyßer, L. T., & Steinberger, J. K. (2020). Scientists' warning on affluence. *Nature Communications, 11*(1), 3107.
World Bank. (2011). *Inclusive green growth: The pathway to sustainable development*. The World Bank.
Wrigley, E. A. (1990). *Continuity, chance and change: The character of the industrial revolution in England*. Cambridge University Press.
Wrigley, E. A. (2013). Energy and the English industrial revolution. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences, 371*(1986), 20110568.
Wrigley, E. A. (2016). *The path to sustained growth: England's transition from an organic economy to an industrial revolution*. Cambridge University Press.
Wu, Y., Zhu, Q. W., & Zhu, B. Z. (2018). Comparisons of decoupling trends of global economic growth and energy consumption between developed and developing countries. *Energy Policy, 116*, 30–38. https://doi.org/10.1016/j.enpol.2018.01.047